

Multi-Task Learning for Bangladeshi Banknote Denomination and Condition Recognition

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Abstract—Reliable banknote recognition under everyday mobile capture conditions is difficult because illumination, background, viewpoint, device characteristics, and physical wear can all vary simultaneously. This study introduces a self-collected Bangladeshi banknote image dataset constructed for two complementary tasks: denomination recognition and note condition classification. The dataset covers seven circulating denominations and three condition labels collected under daylight and low-light conditions, unconstrained backgrounds, and both old and new note categories. Using stratified five-fold cross-validation and training-only offline augmentation, this study evaluated two transfer-learning strategies: independent task-specific networks and a compact MobileNetV3-Large backbone with a shared multi-layer representation and two prediction heads. The independent system achieved the strongest denomination recognition. However, the shared model improves the condition macro-F1 by approximately 3.5 percentage points (pp) and reduces the trainable parameter count by approximately 87%. This improvement is most pronounced for the minority Rare class, suggesting that shared visual features can support the more difficult condition recognition task. Together, the dataset, evaluation protocol, and analysis establish a practical baseline for understanding Bangladeshi banknotes under real-world variations.

Index Terms—Bangladeshi banknotes, denomination recognition, condition classification, multi-task learning, transfer learning, focal loss, assistive technology.

I. INTRODUCTION

Paper currency remains an essential medium for financial transactions, making automatic banknote analysis important for cash-handling systems, vending machines, mobile banking, note-sorting equipment, and assistive technologies for visually impaired users. Unlike laboratory or scanner-based settings, real-world smartphone images often exhibit perspective distortion, illumination changes, shadows, background clutter, occlusions, folds, and device-dependent color variations. Such factors substantially increase the intra-class variation and make reliable recognition considerably more challenging.

Research on Bangladeshi banknote recognition has evolved from handcrafted feature-based methods and conventional neural networks to lightweight deep learning models suitable for mobile deployment [1–5]. Recently, several datasets have been introduced for denomination recognition, banknote detection, and counterfeit analysis [6, 9, 10]. However, each of these datasets is designed to support only one such task in isolation, and none treats the physical condition of banknotes as an explicit learning objective. The condition is instead left as an uncontrolled source of appearance variation.

This study addresses this gap using a real-world dual-label Bangladeshi banknote dataset and evaluates independent and shared learning strategies. Each image contained

both denomination and condition labels, allowing the same visual observation to support two related recognition tasks. The dataset was collected using multiple smartphone cameras across indoor and outdoor environments, daylight and low-light conditions, different backgrounds, and old and new note categories. The final dataset contained 4,260 original images and covered BDT denominations of 10, 20, 50, 100, 200, 500, and 1000.

This study further compared the two practical learning strategies. The independent baseline models were trained separately: ResNet-50 for denomination recognition [20] and MobileNetV3-Large for condition classification [21]. In contrast, the proposed approach adopts a hard parameter-sharing multi-task learning architecture [16], where a shared MobileNetV3-Large backbone is followed by task-specific prediction. Both approaches used ImageNet pre-training [22], class-balanced focal loss [18], AdamW optimization [19], stratified five-fold cross-validation, and data augmentation applied exclusively to the training split. Therefore, this study makes four main contributions. First, this study introduced a self-collected Bangladeshi banknote dataset with joint denomination and condition labels captured under multiple real-world variations. Second, a reproducible preprocessing and stratified cross-validation protocol that preserves the denomination condition distribution across folds while preventing augmented validation samples from entering training is proposed. Third, independent task-specific transfer learning models with a compact shared multi-task learning architecture for joint recognition. Finally, a class-level analysis shows that shared learning improves the more difficult condition recognition task, particularly for the minority Rare class, while maintaining strong denomination recognition performance.

II. LITERATURE REVIEW

A. Banknote Recognition and Assistive Systems

Automatic banknote recognition has evolved from handcrafted feature-based methods to deep learning using RGB cameras and other sensing modalities. The major challenges include illumination changes, orientation variation, physical degradation, and complex backgrounds [11]. Early Bangladeshi systems relied on handcrafted features and conventional neural network classifiers [1]. Recent studies have adopted transfer learning with lightweight CNN architectures such as MobileNet, NASNetMobile, and ResNet for denomination recognition and mobile assistive applications [2–5]. These advances have also facilitated the development of integrated assistive systems that combine banknote recognition

with navigation, object detection, and emergency support features for visually impaired users [14]. However, existing studies primarily focus on denomination recognition, while banknote condition is generally treated as a source of appearance variation rather than an explicit prediction target.

B. Bangladeshi Banknote Datasets

Several public datasets have been used for Bangladeshi banknote research. NSTU-BDTAKA targets denomination detection and recognition [6], BanglaTaka provides diverse smartphone-captured images [9], and JaalTaka focuses on counterfeit detection [10]. Despite these contributions, no existing dataset provides joint denomination and condition annotations for supervised learning.

C. Banknote Condition Assessment

Banknote condition assessment has mainly been studied for automated cash sorting systems using controlled imaging and specialized sensors. Kwon *et al.* proposed fitness classification using visible light and near infrared images [12], while Pham *et al.* developed deep-learning methods for banknote fitness and joint type–fitness recognition [13, 15]. However, these approaches are designed for controlled acquisition rather than for unconstrained smartphone images.

D. Multi-Task Learning and Imbalanced Classification

Multi-task learning (MTL) enables related tasks to share feature representations while maintaining task-specific outputs [16, 27]. This study adopts hard parameter sharing with equal task weighting to evaluate the effect of shared representation learning. Because the condition dataset is imbalanced, focal loss [18] is used to improve learning for minority classes, while AdamW, batch normalization, and dropout are employed to enhance optimization and generalization.

III. METHODOLOGY

A. Problem Formulation

This study aims to jointly classify the denomination and physical condition of Bangladeshi banknotes. Given an input image x , the denomination task predicts one of seven denomination classes, whereas the condition task predicts one of three condition categories.

Let y_d and y_c denote the ground truth denomination and condition labels, respectively. The independent baseline trains each task using separate networks, whereas the proposed model learns a shared feature representation as follows:

$$z = f_\theta(x), \quad (1)$$

where f_θ denotes the shared feature extractor and z is the learned feature vector. The task specific predictions are given by

$$\hat{y}_d = g_{\phi_d}(z), \quad (2)$$

$$\hat{y}_c = g_{\phi_c}(z), \quad (3)$$

where g_{ϕ_d} and g_{ϕ_c} are the denomination and condition classification heads. Although both tasks share visual characteristics,

such as color, texture, and printed structures, they require separate prediction heads because their classification objectives differ.

B. Independent Dual-Model Baseline

The independent baseline employs two transfer learning models that are trained separately without sharing feature representations. Denomination classification uses an ImageNet-pretrained ResNet-50 [20, 22], where the final classification layer is replaced with a seven-class fully connected layer corresponding to the denomination of Bangladeshi banknotes.

Condition classification uses an ImageNet-pretrained MobileNetV3-Large [21, 22] with its final classification layer replaced by a three-class output representing the Fresh, Mint, and Rare categories. The two models were optimized independently without exchanging intermediate features or predictions. Together, they contained approximately 27.7 million trainable parameters. The overall architecture is shown in Fig. 1.

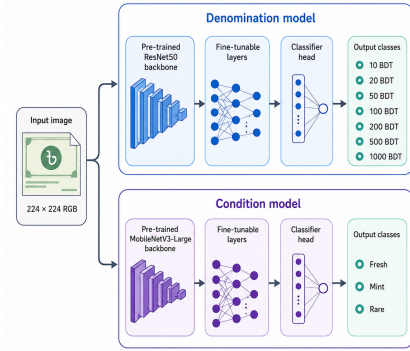


Fig. 1: Independent dual-model baseline using ResNet-50 and MobileNetV3-Large for separate denomination and condition classification.

C. Shared Multi-Task Learning Architecture

The proposed model adopts a hard parameter-sharing multi-task learning (MTL) architecture [16], where a single feature extractor is shared between the denomination and condition classifications. The architecture consists of a shared MobileNetV3-Large backbone, shared multilayer representation, and two task-specific prediction heads.

An ImageNet-pretrained MobileNetV3-Large [21, 22] was used as the shared backbone. After adaptive average pooling, the extracted 960-dimensional feature vector was passed through a multilayer perceptron with the following structure:

$$960 \rightarrow 512 \rightarrow 256. \quad (4)$$

Each fully connected layer is followed by batch normalization [7], ReLU activation, and dropout [8] with a probability of 0.35. The resulting 256-dimensional representation is then fed to separate denomination and condition classification heads as follows:

$$x \rightarrow f_\theta(x) \rightarrow z \rightarrow \begin{cases} g_{\phi_d}(z), \\ g_{\phi_c}(z). \end{cases} \quad (5)$$

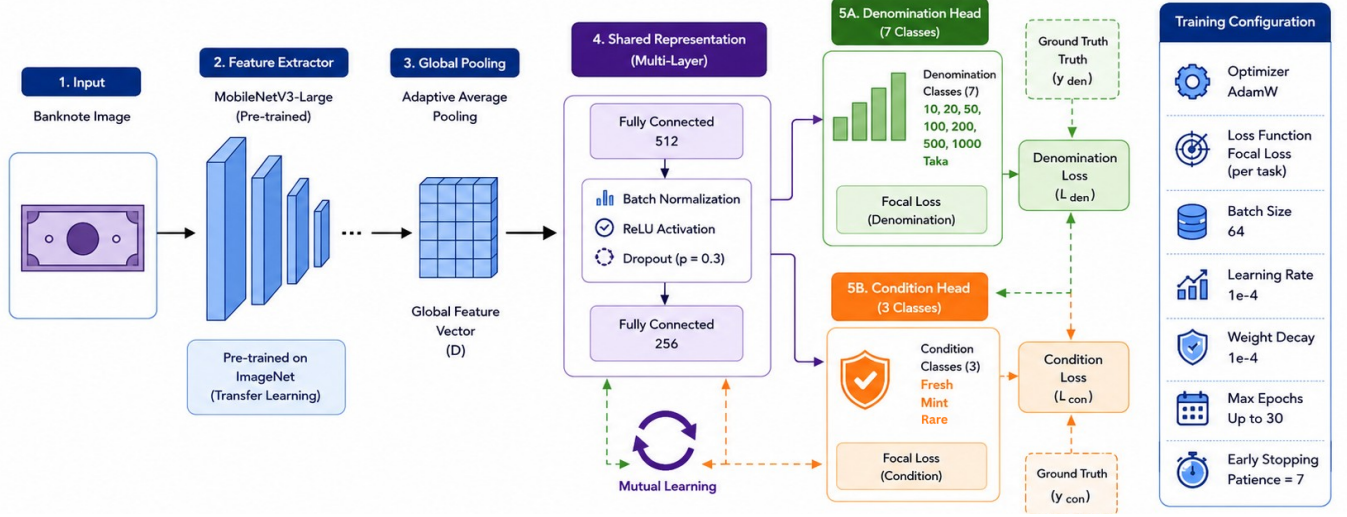


Fig. 2: Shared hard-parameter multi-task architecture using a MobileNetV3-Large backbone for joint feature learning and task-specific prediction heads.

The proposed architecture 2 contains approximately 3.6 million trainable parameters, representing an 87% reduction compared with that of the independent baseline. Unlike deep mutual learning, which employs multiple peer networks with prediction consistency objectives [23], the proposed model uses a single shared backbone with two supervised, task-specific heads.

D. Dataset Preparation and Preprocessing

The proposed dataset comprises 4,260 RGB images of Bangladeshi banknotes captured using multiple smartphone cameras under diverse real-world conditions, including indoor and outdoor environments, daylight and low-light illumination, varying backgrounds, and viewpoints and orientations. Each image is annotated with both a denomination label (BDT 10, 20, 50, 100, 200, 500, or 1000) and a condition label (Fresh, Mint, or Rare).

The condition classes are imbalanced, consisting of 2,329 Fresh, 1,501 Mint, and 430 Rare samples. In total, 20 denomination-condition combinations are represented, with no Rare samples for the BDT 200 denomination.

All images were resized to 224×224 pixels and normalized using ImageNet statistics [22]:

$$\mu = (0.485, 0.456, 0.406), \quad (6)$$

$$\sigma = (0.229, 0.224, 0.225). \quad (7)$$

Data augmentation was applied exclusively to the training split using random rotations, brightness and contrast adjustments, Gaussian noise, and horizontal flipping. Restricting augmentation to the training data prevents information from leaking into the validation set.



Fig. 3: Representative samples from the proposed dataset showing variations in denomination, environment, lighting, background and note condition.

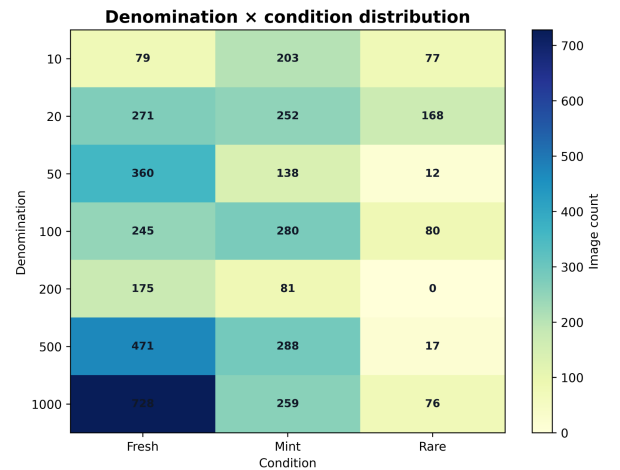
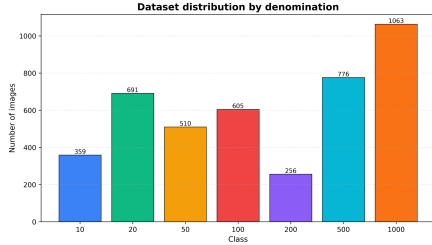
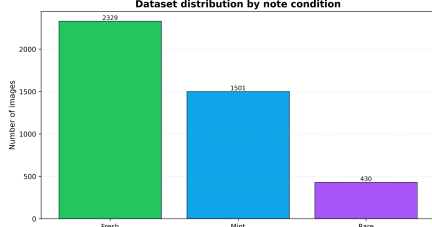


Fig. 5: Joint distribution of denomination and condition labels, showing 20 valid combinations and no BDT 200 samples in the Rare class.



(a) Denomination distribution.



(b) Condition distribution.

Fig. 4: Distribution of the 4,260 original banknote images by denomination and condition.

E. Loss Function

To mitigate class imbalance, particularly for condition classification, focal loss [18] is adopted instead of the standard cross-entropy:

$$\mathcal{L}_{\text{focal}} = -\alpha_t(1 - p_t)^\gamma \log(p_t), \quad (8)$$

where p_t is the predicted probability of the target class, α_t is the class balancing factor, and γ is the focusing parameter. Throughout this study, $\gamma = 2.0$. For the multi-task model, the overall objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{denomination}} + \mathcal{L}_{\text{condition}}, \quad (9)$$

where $\mathcal{L}_{\text{denomination}}$ and $\mathcal{L}_{\text{condition}}$ are computed using Eq. (8) applied to each task's logits.

F. Optimisation and Training Strategy

Both models were initialized with ImageNet-pretrained weights and optimized using AdamW [19] with a learning rate of 1×10^{-4} and weight decay of 1×10^{-4} . The performance is evaluated using stratified five-fold cross-validation, preserving the joint denomination-condition distribution across folds. The checkpoint with the highest average denomination and condition macro-F1 score is retained. Early stopping with a patience of seven epochs is applied, with a maximum of 10 epochs for the independent baseline and 30 epochs for the shared multi-task model.

IV. EXPERIMENTS AND RESULTS

A. Evaluation Metrics

Performance is evaluated using the accuracy and macro-F1 score for both denomination and condition classification.

Accuracy is defined as:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Samples}}, \quad (10)$$

where the class-wise F1 score is

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (11)$$

Macro-F1 is emphasized because it provides a balanced evaluation of class imbalance. Note that the cross-validation protocol has an important caveat affecting the interpretation of these results, which is discussed in Section V-A. The results are reported as the mean and standard deviation across the five folds, whereas out-of-fold predictions are aggregated to generate the final confusion matrices and class-wise analyses.

B. Overall Performance Comparison

Table I compares the performance of the independent dual-model baseline and the proposed shared multitask model.

TABLE I: Five-Fold Performance Comparison

Method	D-Acc	D-F1	C-Acc	C-F1	Parameters
Independent	97.91 ± 0.51	97.97 ± 0.66	81.08 ± 0.82	75.01 ± 1.17	27.73M
Shared	97.11 ± 0.24	97.13 ± 0.31	83.17 ± 0.98	78.55 ± 1.86	3.60M
Difference	0.80	0.84	2.09	3.54	87%

TABLE II: Mean five-fold performance comparison between the independent and shared models (%).

Metric	Independent	Shared
Denomination Accuracy	97.91	97.11
Denomination Macro-F1	97.97	97.13
Condition Accuracy	81.08	83.17
Condition Macro-F1	75.01	78.55

The independent model achieved the highest denomination performance (97.91% accuracy), whereas the shared model improved condition classification, increasing accuracy and macro-F1 by 2.09 and 3.54 percentage points, respectively, while reducing trainable parameters by 87% (Table II).

C. Fold-Wise Validation Results

Table III summarizes these five-fold results. The shared model consistently achieved a higher condition macro-F1 across all folds.

TABLE III: Fold-wise Validation Performance (%)

Method	Fold	D-Acc	D-F1	C-Acc	C-F1
Independent	0	98.59	98.72	80.05	74.92
	1	97.77	97.94	81.22	74.17
	2	98.12	98.37	80.52	74.88
	3	97.89	97.87	81.46	74.09
	4	97.18	96.96	82.16	77.00
Shared	0	97.30	97.42	82.28	79.12
	1	97.18	97.27	82.86	78.00
	2	97.30	97.32	84.15	79.56
	3	97.07	96.97	82.28	75.62
	4	96.71	96.66	84.27	80.45

D. Denomination Recognition Performance

Both models achieved high denomination recognition accuracy. The independent model correctly classified 4,171 of 4,260 images, whereas the shared model correctly classified 4,137. The class-wise F1 scores are shown in Table IV.

TABLE IV: Class-wise Denomination F1 Score (%)

Method	10	20	50	100	200	500	1000	Macro
Independent	98.21	96.90	98.43	97.37	98.82	97.36	98.72	97.97
Shared	98.18	95.68	97.73	97.21	96.90	96.01	98.21	97.13

The shared model showed slightly higher confusion between visually similar denominations, mainly BDT 20 and BDT 500 (Fig. 4a).

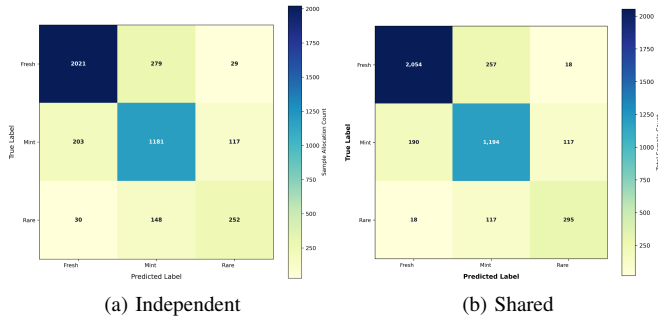


Fig. 6: Aggregate out-of-fold confusion matrices for condition classification. The shared model reduces Fresh–Mint confusion and improves recognition of the minority Rare class.

E. Condition Classification Performance

Condition classification is more challenging because of the class imbalance. Table V shows that the shared model improves all classes, with the Rare-class F1 increasing from 60.87% to 68.60%.

TABLE V: Condition Classification Performance (%)

Method	Class	Precision	Recall	F1
Independent	Fresh	89.66	86.78	88.20
	Mint	73.45	78.68	75.97
	Rare	63.32	58.60	60.87
Shared	Fresh	90.80	88.19	89.48
	Mint	76.15	79.55	77.81
	Rare	68.60	68.60	68.60

The confusion matrices in Fig. 6 show reduced Fresh–Mint confusion and better Rare-class recognition.

F. Model Efficiency Analysis

As shown in Table I, the shared architecture reduces the number of trainable parameters from 27.73M to 3.60M (87%) while improving condition classification, making it suitable for mobile deployment.

V. LIMITATIONS AND FUTURE WORK

A. Limitations

Several limitations should be considered. First, image-level stratified cross-validation was used. Although augmented vali-

dation images were excluded from training, images of the same banknote or capture session may appear in different folds, potentially introducing identity-level information leakage. Future work should adopt grouped cross-validation based on banknote identity or capture session.

Second, the dataset is imbalanced, with substantially fewer Rare samples than Fresh and Mint samples (Figs. 4b and 5). Moreover, no BDT 200 banknotes belong to the Rare class, limiting the evaluation of all denomination–condition combinations.

Third, Fresh and Mint labels are inherently subjective. Since each image was annotated once, inter-annotator agreement could not be assessed. Future datasets should include multiple annotators, clear annotation guidelines, and agreement analysis.

Finally, the independent and shared models used different training settings. Therefore, Table I compares the best-performing implementations rather than a strictly controlled architectural comparison.

B. Future Work

Future work will focus on expanding the dataset with more balanced denomination condition combinations and additional real-world variations to improve the generalization of the model. The learning framework can also be extended by investigating adaptive multitask optimization strategies, such as uncertainty-based loss weighting [17] and GradNorm [24], to better balance the two tasks.

The proposed architecture will be further evaluated on mobile devices by measuring the inference latency, memory usage, and energy consumption. A complete assistive system could additionally integrate banknote detection, explainability techniques such as Grad-CAM [25], and support for related tasks, including counterfeit detection, severe damage assessment, and security feature recognition.

VI. CONCLUSION

This study presents a real-world Bangladeshi banknote recognition framework for joint denomination and condition classification. A new dataset containing 4,260 smartphone-captured images with dual denominationcondition labels was introduced, covering seven denominations and three condition categories under diverse real-world conditions.

Experimental results showed that the independent model achieved the highest denomination recognition accuracy (97.91%), whereas the proposed shared multitask model improved condition classification, increasing macro-F1 from 75.01% to 78.55%, while reducing the number of trainable parameters by 87% (27.73M to 3.60M).

Overall, the results demonstrate that shared feature learning provides an effective trade-off between accuracy and model efficiency, making the proposed approach a practical foundation for mobile banknote recognition, assistive technologies, and future condition-aware currency analysis.

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